1. **What is React, and why is it used in web development?**
   React is a JavaScript library for building UI. It helps make websites fast, reusable, and easier to manage with components.

2. **What is JSX? Mention one advantage of using JSX.**
   JSX is a syntax that lets us write HTML-like code inside JavaScript. One big advantage is it makes the code easier to read and write.

3. **Explain the difference between Props and State in React.**
   Props are data passed from parent to child component, and they are read-only. State is local data inside a component and it can change.

4. **What is the purpose of the useState() hook? Give a simple example.**
   useState() is used to store and update data in a component.
   Example:
   const [count, setCount] = useState(0);
   Here count starts with 0 and setCount() updates it.

5. **What is the useEffect() hook, and when is it commonly used?**
   useEffect() runs code after the component renders. Mostly used for API calls, fetching data, timers, or event listeners.

6. **What is the Virtual DOM? How does it improve React application performance?**
   Virtual DOM is a copy of the real DOM. React checks the changes in Virtual DOM first and updates only the changed parts, so it's faster.

7. **Why is the key prop important when rendering lists in React?**
   The key helps React identify which item changed, added, or removed. This makes rendering faster and avoids unnecessary updates.

8. **Explain the purpose of the useRef() hook with an example use case.**
   useRef() keeps a value without causing re-render. Common use is accessing an input field.
   Example:
   const inputRef = useRef();
   inputRef.current.focus();

9. **What is the difference between useMemo() and useCallback()?**
   useMemo() remembers a calculated value. useCallback() remembers a function. Both help improve performance.

10. **What is the Context API, and when would you use it instead of passing props through multiple components?**
    Context API is used to share data between many components without passing props again and again. It's useful for things like user login info, theme, or language settings.